

EC331: Empirical specification

Mahima Sabberwal

Research question

- **RQ:** What were the long-run effects of 1991 trade liberalisation reforms (through looking at tariff reductions) on local labour market outcomes in India?
- Outcomes of interest: Employment (current)
Aim to also look at: fertility, age of women at marriage, total length of roads in a region
- Two data points: 1991 and 2001

1 Specification

OLS (not estimated - expect employment shares to react to industry tariff changes, and evidence of tariff assignment after 1997 being correlated with industry characteristics - IMF imposed tariff reductions between 1991 - 1997):

$$Y_{dt} = \alpha + \beta Protection_{dt} + \gamma_d + \lambda_t + \epsilon_{dt}$$

Let w be the employment share for industry j in region d at time t . Employment share for industry j = Number of workers in j /total number of workers

$Protection_{dt} = \sum_j w_{dj} \cdot Tariff_{jt}$ by identity; the net exposure to tariffs in each region

SSIV regression - main reduced form specification:

$$Y_{dt} = \alpha + \beta \sum_j w_{dj,1991} \cdot Tariff_{j1991,1997} + \gamma_d + \lambda_t + \epsilon_{dt}$$

I have instrumented for $Protection_{dt}$ with the sum of: baseline 1991 employment share for a particular industry in region d interacted with the national tariff for products in that industry (averaged) in 1991 or 1997 (for 2001 observations), for all industries. i.e. my treatment and instrument values are identical for 1991.

- Region FEs accounting for ‘incomplete shares’ (‘other service’ industry being assigned a tariff of 0)
- Clustered standard errors at the composite region level, but aim to compute exposure-robust SEs (ssaggregate package, following Borusyak, Hull and Jaravel)
- LHS employment is currently the share of main workers of the population. At recommendation of my supervisor, will also estimate with log(number of employed workers) on the LHS, including log(population) on the RHS.

Considering the effects of gender-specific exposure to trade on employment outcomes, by multiplying industry shares with gender component:

$$Y_{dt} = \alpha + \beta_1 \sum_j w_{fdj,1991} \cdot \text{Tariff}_{j1991,1997} + \beta_2 \sum_j w_{mdj,1991} \cdot \text{Tariff}_{j1991,1997} + \gamma_d + \lambda_t + \epsilon_{dt}$$

(I now consider region-by-industry-by-gender employment shares - considering the proportion of female f and male m workers in each industry)

2 Concerns

- Potentially insufficient variation in employment within districts over time (employment mean overall: 0.404, standard deviation within T: 0.0369)
- Checking for/validating the assumption that tariffs are as-good-as-randomly assigned to industries (i.e. tariffs are not systematically different in industries which are concentrated in particular regions exposed to labour supply shocks). Ideas: descriptively having a map showing treatment/instrument exposure at the region level. I do not have employment data prior to 1991 to regress on lagged outcomes.